

Experiment 07: To develop an android application that uses GPS Location Information.

PART A

A.1 Aim: To develop an android application that uses GPS Location Information.

A.2 Objectives: To introduce students with various tools like Android Studio, NS2, Wire-shark, Cisco packet tracer, WAP supported browser etc.

A.3 Outcome: After successful completion of this experiment students will be able to develop an android application that uses GPS Location Information. **A.4 Theory:**

SOFTWARE:

- Android Studio
- The Android SDK (Starter Package)
- Gradle
- Java Development Kit (JDK) 5

DESCRIPTION:

1. Open android studio and select new android project .
2. Give project name and select next
3. Choose the android version.
4. Enter the package name. package name must be two word separated by comma and click finish
5. Go to package explorer in the left hand side and select our project.
6. Go to res folder and select layout. Double click the main.xml file
7. Now you can see the Graphics layout window.

INPUT:

MAINACTIVITY.JAVA

```
package com.gpslocation;

//import android.R; import android.view.View;
import android.widget.Button; import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity; import android.os.Bundle;

public class MainActivity extends AppCompatActivity{

    /** Called when the activity is first created. */ Button
    btnShowLocation; GPSTrace gps;

    @Override
    public void onCreate(Bundle savedInstanceState) { super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        btnShowLocation=(Button)findViewById(R.id.show_Location);
        btnShowLocation.setOnClickListener(new View.OnClickListener() {

            @Override
            public void onClick(View v) {
                // TODO Auto-generated method stub gps=new
                GPSTrace(MainActivity.this);
                if(gps.canGetLocation()) { double
                latitude=gps.getLatitude(); double
                longitude=gps.getLongitude();
                Toast.makeText(getApplicationContext(),"Your Location is
                \nLat:"+latitude+"\nLong:"+longitude, Toast.LENGTH_LONG).show();
                }
                else { gps.showSettingAlert();
                }

            }

        });
    }
}
```

ACTIVITY_MAIN.XML

```
<?xml version="1.0" encoding="utf-8"?>
```

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```
<RelativeLayoutxmlns:android="http://schemas.android.com/apk/res/android"andro  
id:id="@+id/relativeLayout1" android:layout_width="match_parent"  
android:layout_height="match_parent">  
  
    <Button android:id="@+id/show_Location"  
    android:layout_width="wrap_content"  
    android:layout_height="wrap_content"  
    android:text="Show_Location"  
    android:layout_centerVertical="true"  
    android:layout_centerHorizontal="true" />  
  
</RelativeLayout>
```

GPSTRACE.JAVA

```
package com.gpslocation;
```

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```
import android.app.AlertDialog; import
android.app.Service; import android.content.Context;
import android.content.DialogInterface; import
android.content.Intent; import
android.location.Location; import
android.location.LocationListener; import
android.location.LocationManager; import
android.os.Bundle; import android.os.IBinder;
import android.provider.Settings;
```

```
public class GPSTrace extends Service implements LocationListener { private final Context
    context; boolean isGPSEnabled=false; boolean canGetLocation=false;
    boolean isNetworkEnabled=false; Location location;
    double latitude; double longitude;
    private static final long MIN_DISTANCE_CHANGE_FOR_UPDATES=10; private static
    final long MIN_TIME_BW_UPDATES=1000*60*1; protected
    LocationManager locationManager; public
    GPSTrace(Context context) {
        this.context=context; getLocation();
    }
    public Location getLocation() { try { locationManager=(LocationManager)
context.getSystemService(LOCATION_SERVICE);

isGPSEnabled=locationManager.isProviderEnabled(LocationManager.GPS_PROVIDER
);

isNetworkEnabled=locationManager.isProviderEnabled(LocationManager.NETWORK_PROVIDER);

        if(!isGPSEnabled && !isNetworkEnabled) {
        }
        else { this.canGetLocation=true;
            if(isNetworkEnabled) {

locationManager.requestLocationUpdates(                LocationManager.NETWORK_PROVIDER,
MIN_TIME_BW_UPDATES, MIN_DISTANCE_CHANGE_FOR_UPDATES, this);
            }
            if(locationManager!=null) {

location=locationManager.getLastKnownLocation(LocationManager.NETWORK_PROVIDER);
                if(location !=null) { latitude=location.getLatitude();
longitude=location.getLongitude();
                }
            }
        }

        if(isGPSEnabled) { if(location==null)
        {
```

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```
locationManager.requestLocationUpdates(LocationManager.GPS_PROVIDER,MIN_TIME,BW_UPDATES,MIN_DISTANCE_CHANGE_FOR_UPDATES, this); if(locationManager!=null) {
```

```
location=locationManager.getLastKnownLocation(LocationManager.GPS_PROVIDER)
```

```
;
```

```
if(location!=null) { latitude=location.getLatitude();
```

```
longitude=location.getLongitude();
```

```
}
```

```
}
```

```
}
```

```
}
```

```
}
```

```
catch(Exception e) {
```

```
e.printStackTrace();
```

```
}
```

```
return location;
```

```
}
```

```
public void stopUsingGPS() { if(locationManager!=null) {
```

```
locationManager.removeUpdates(GPSTrace.this);
```

```
}
```

```
}
```

```
public double getLatitude() {
```

```
if(location!=null) {
```

```
latitude=location.getLatitude();
```

```
}
```

```
return latitude;
```

```
}
```

```
public double getLongitude() {
```

```
if(location!=null) {
```

```
longitude=location.getLatitude();
```

```
}
```

```
return longitude;
```

```
}
```

```
public boolean canGetLocation() { return this.canGetLocation;
```

```
}
```

```
public void showSettingAlert() {
```

```
AlertDialog.Builder alertDialog= new AlertDialog.Builder(context);
```



```
        alertDialog.setTitle("GPS is settings"); alertDialog.setMessage("GPS is not enabled.Do  
        you want to go to  
        setting menu?");  
        alertDialog.setPositiveButton("settings", new  
        DialogInterface.OnClickListener() {  
            @Override  
            public void onClick(DialogInterface dialog, int which) { Intent intent= new  
            Intent(Settings.ACTION_LOCATION_SOURCE_SETTINGS);  
                context.startActivity(intent);  
            }  
        });  
        alertDialog.setNegativeButton("cancel", new  
        DialogInterface.OnClickListener() {  
            @Override  
            public void onClick(DialogInterface dialog, int which) {  
                // TODO Auto-generated method stub dialog.cancel();  
            }  
        });  
        alertDialog.show();  
    }  
  
    @Override  
    public void onLocationChanged(Location location) {  
        // TODO Auto-generated method stub  
    }  
  
    @Override  
    public void onProviderDisabled(String provider) {  
        // TODO Auto-generated method stub  
    }  
  
    @Override  
    public void onProviderEnabled(String provider) {  
        // TODO Auto-generated method stub  
    }  
  
    @Override  
    public void onStatusChanged(String provider, int status, Bundle extras)  
{  
        // TODO Auto-generated method stub  
    }  
  
    @Override  
    public IBinder onBind(Intent intent) {  
        // TODO Auto-generated method stub return  
        null;  
    }  
}
```

```
<?xml version="1.0" encoding="utf-8"?>
```

<manifest

```
xmlns:android="http://schemas.android.com/apk/res/android" package="com.gpsl
ocation">
```

<application

```
android:allowBackup="true"
android:icon="@mipmap/ic_launcher"
android:label="@string/app_name"
android:roundIcon="@mipmap/ic_launcher_round"
android:supportsRtl="true"
android:theme="@style/Theme.GPSLocation">
```

<activity

```
android:name=".MainActivity"
android:exported="true">
```

<intent-filter>

```
<action android:name="android.intent.action.MAIN" />
```

```
<category android:name="android.intent.category.LAUNCHER"/>
```

</intent-filter>

</activity>

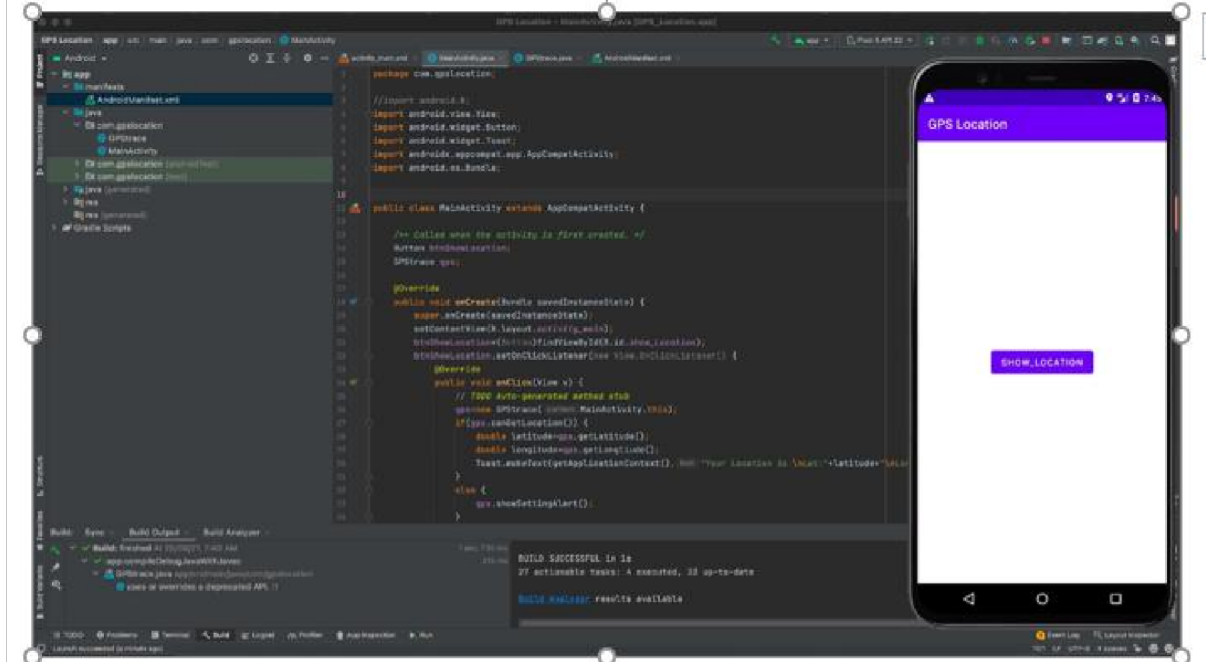
</application>

<uses-permission

```
android:name="android.permission.ACCESS_FINE_LOCATION"/>
```

```
<uses-permission android:name="android.permission.INTERNET"/>
```

</manifest>

[illegible]



SHOW_LOCATION



SHOW_LOCATION

Your Location is
Lat:37.421998333333335
Long:37.421998333333335



PART B

(PART B: TO BE COMPLETED BY STUDENTS)

(Students must submit the soft copy as per following segments within two hours of the practical. The soft copy must be uploaded on the Blackboard or emailed to the concerned lab in charge faculties at the end of the practical in case the there is no Black board access available)

Roll. No. : C-54	Name: Quazi Md Moinuddin
Class : TE Comps C	Batch: C-3
Date of Experiment: 17.03.25	Date of Submission:27.03.25
Grade:	

B.1 Software Code written by student/steps:

AndroidManifest.xml

```
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    package="com.example.gpslocationapp">

    <!-- Permissions for Internet and Location -->
    <uses-permission android:name="android.permission.INTERNET"/>
    <uses-permission android:name="android.permission.ACCESS_FINE_LOCATION"/>
    <uses-permission android:name="android.permission.ACCESS_COARSE_LOCATION"/>
```

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```
<application
    android:allowBackup="true"
    android:icon="@mipmap/ic_launcher"
    android:label="GPS Location App"
    android:roundIcon="@mipmap/ic_launcher_round"
    android:supportRtl="true"
    android:theme="@style/Theme.Material3.Light">

    <!-- Fix: Add android:exported="true" -->
    <activity android:name=".MainActivity"
        android:exported="true">
        <intent-filter>
            <action android:name="android.intent.action.MAIN"/>
            <category android:name="android.intent.category.LAUNCHER"/>
        </intent-filter>
    </activity>

</application>

</manifest>
```

MainActivity.java

```
package com.example.gpslocationapp;

import android.location.Address;
import android.location.Geocoder;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;
import java.io.IOException;
import java.util.List;
import java.util.Locale;

public class MainActivity extends AppCompatActivity {

    private EditText locationInput;
    private Button findButton;
    private TextView resultText;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
```

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```
        setContentView(R.layout.activity_main);

        locationInput = findViewById(R.id.locationInput);
        findButton = findViewById(R.id.findButton);
        resultText = findViewById(R.id.resultText);

        findButton.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String locationName =
locationInput.getText().toString().trim();
                if (!locationName.isEmpty()) {
                    getCoordinates(locationName);
                } else {
                    Toast.makeText(MainActivity.this, "Please enter a
location!", Toast.LENGTH_SHORT).show();
                }
            }
        });

        private void getCoordinates(String location) {
            Geocoder geocoder = new Geocoder(this, Locale.getDefault());
            try {
                List<Address> addresses = geocoder.getFromLocationName(location, 1);
                if (addresses != null && !addresses.isEmpty()) {
                    Address address = addresses.get(0);
                    double latitude = address.getLatitude();
                    double longitude = address.getLongitude();

                    resultText.setText("Coordinates:\nLatitude: " + latitude +
"\nLongitude: " + longitude);
                } else {
                    resultText.setText("Location not found!");
                }
            } catch (IOException e) {
                Toast.makeText(this, "Error fetching location!",
Toast.LENGTH_SHORT).show();
                e.printStackTrace();
            }
        }
    }
}
```

Activity_main.xml

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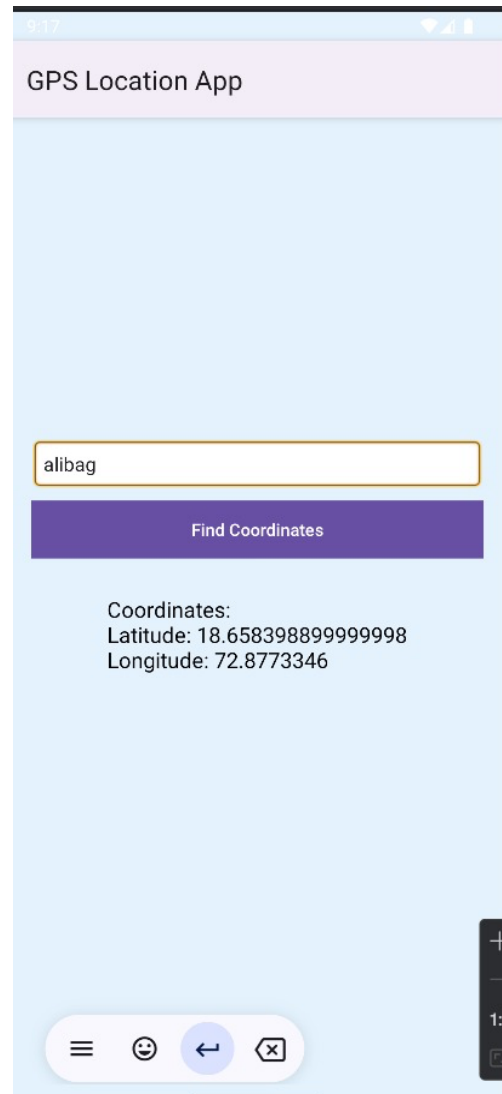
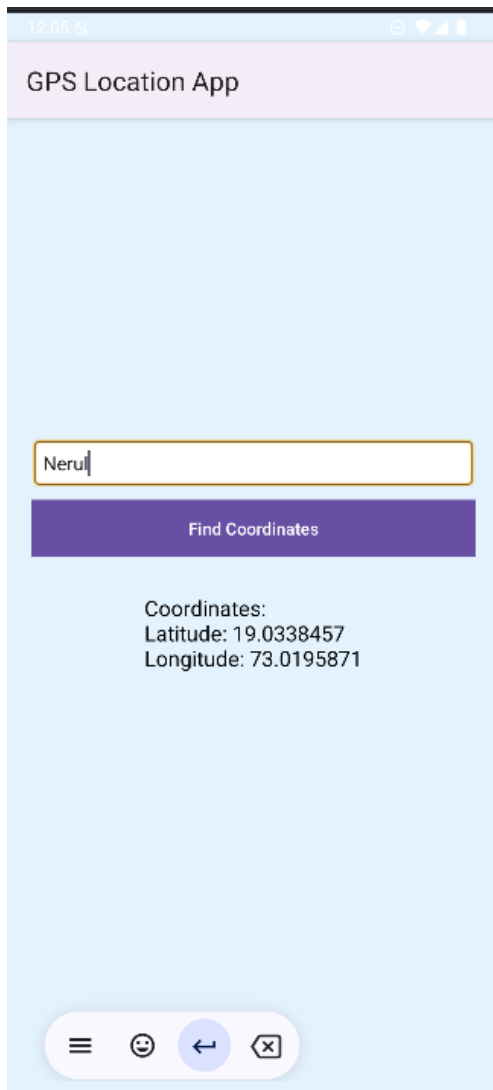
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp"
    android:background="#E3F2FD">

    <EditText
        android:id="@+id/locationInput"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter location name"
        android:textSize="16sp"
        android:padding="10dp"
        android:background="@android:drawable/editbox_background"/>

    <Button
        android:id="@+id/findButton"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Find Coordinates"
        android:background="#2196F3"
        android:textColor="#FFFFFF"
        android:padding="10dp"
        android:layout_marginTop="10dp"/>

    <TextView
        android:id="@+id/resultText"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Coordinates will appear here"
        android:textSize="18sp"
        android:textColor="#000000"
        android:padding="10dp"
        android:layout_marginTop="20dp"/>
</LinearLayout>
```

B.2 Input and Output:



B.3 Observations and learning:

The application effectively accesses the device's GPS module and handles location data. It maintains accuracy under open-sky conditions, but performance may degrade in areas with poor satellite visibility, such as indoors or dense urban environments. Additionally, permissions management and battery consumption are important considerations for real-world applications.

B.4 Conclusion:

The Android application utilizing GPS location information successfully retrieves and displays the user's current geographical coordinates. By leveraging location services, the app provides real-

time updates, ensuring accurate tracking. It demonstrates the practical use of GPS in mobile applications for location-based services like navigation, ride-hailing, and location sharing.

B.5 Question of Curiosity

1) Explain different steps required to build up this GPS project?

Steps to Build a GPS-based Android Application

1. Set Up Android Studio:

- Create a new Android project using **Empty Activity**.

2. Add Permissions:

- In `AndroidManifest.xml`, add location permissions using `ACCESS_FINE_LOCATION` and `ACCESS_COARSE_LOCATION`.

3. Add Dependencies:

- Add Google Play Services Location API in `build.gradle`.

4. Design UI:

- Create a simple interface with a **Button** to get location and a **TextView** to display coordinates.

5. Implement GPS Functionality:

- Use `FusedLocationProviderClient` to fetch the user's current location.

6. Request Permissions:

- Handle location permission requests using `ActivityCompat.requestPermissions()`.

7. Display Location:

- Once permission is granted, display latitude and longitude in the `TextView`.

8. Test the App:

- Run the app on a physical device or emulator with GPS enabled.